

Parts Inquiry & SAP Business One PO API Integration

Operational Context, Komatsu PDX Split Rules, and Unified SAP B1 Service Layer Purchase Order Creation

TARGET AUDIENCE	PREPARED BY	CORE SYSTEMS	DOCUMENT VERSION
ABS (SAP B1 Technical Team)	Motasem Ghanem (EQP Ops)	EQP Portal • Komatsu PDX • SAP B1	v2.0 (Unified SAP PO Creation RFC)

1. Executive Summary: All Orders Are Identical to SAP

Dar Al Hai is the authorized Komatsu equipment operator and service organization managing an extensive fleet of excavators, bulldozers, and heavy earthmoving machinery across major infrastructure projects in Kuwait. To maintain operational readiness and rapidly replenish replacement components, the Maintenance & Spare Parts department orders genuine parts directly from **Komatsu Middle East (KME / PDX Portal)**.

To ensure seamless accounting integration, the EQP maintenance system connects directly with **SAP Business One** to create corresponding **Purchase Orders (POs)** for Vendor **v000006 (Komatsu Middle East)** in US Dollars (USD).

Core Operating Principle for the ABS Team:
To SAP Business One, all orders are identical standard Purchase Orders.
Whether a sub-order originated from an Emergency Order (EO) or a Normal Stock Order (DS / SO), the SAP integration structure and process do not change:

- Automatic Pricing from Komatsu:** The unit price generated in Komatsu PDX already reflects the final price. When an order is placed on an Emergency basis (EO), Komatsu PDX automatically adjusts the unit price directly on the quotation. When placed as Daily Stock (DS), it applies normal catalogue rates.
- Direct Price Passthrough:** Neither SAP nor the API integration middleware needs to calculate surcharges, premiums, or special freight formulas. SAP simply records the exact confirmed unit price received from Komatsu.
- Uniform PO Structure:** Every confirmed Komatsu Sales Order (SO) creates one standard Purchase Order in SAP B1 using the exact same API payload schema.

2. End-to-End System Architecture Flow

The workflow connects field requisitions to SAP Business One through a clean, automated API sequence:



3. The Operational Need: Why Orders Are Split (Komatsu Portal Constraint)

From an ERP perspective, purchasing 50 filters would normally be handled as a single PO line. However, the **Komatsu Middle East PDX Portal** enforces strict manufacturer routing and anti-hoarding policies that require the EQP system to split requests into multiple sub-orders:

- **Channel Distinction on Komatsu PDX:**
 - **Normal Stock Orders (DS Daily Stock / SO Stock Order):** Fulfilled up to available stock in KME Dubai local warehouse. Komatsu allows bulk quantities with **no machine serial number or customer details**.
 - **Emergency Orders (EO):** Used for shortages and factory backorders. Komatsu strictly mandates that every emergency line item must be validated against a genuine machine model (e.g. PC500LC-10R) and Kuwait chassis serial number (e.g. 100433).
- **Anti-Hoarding Line Item Quotas (Emergency Channel):** Komatsu caps emergency breakdown quantities per machine (e.g. max 20 filters per machine). Submitting 50 units against a single excavator is rejected by portal validation.
- **Automated Komatsu Pricing:** When Komatsu generates an Emergency quotation, it automatically applies its emergency pricing structure to the unit price. When generating a Daily Stock quotation, it applies base rates.
- **Sequential Reference Numbering:** Each dispatched Komatsu sub-order receives a continuous sequential distributor reference (e.g. R229/2026 , R230/2026 , R231/2026).

Summary for SAP Team: Order Splitting is Strictly an External Komatsu Operational Step

The splitting logic and machine allocation exist solely to comply with Komatsu's manufacturer portal rules. **Once Komatsu approves and confirms the sub-orders into official Komatsu Sales Orders (SOs), each SO maps directly 1:1 into a standard SAP Business One Purchase Order.**

4. Illustrative Example: 50 Units of Fuel Filter (Part 600-311-6742)

When a maintenance requisition for 50 units is processed:

Requisition: 50 EA • Fuel Filter (600-311-6742)

Komatsu PDX Stock Availability: KME Stock: 10 | Shortage: 40

1. Komatsu Sub-Order #1 (DS Stock) — Ref R229/2026 : 10 units fulfilled from Dubai stock → Converts to Komatsu SO #0000278046 at confirmed catalogue price → Creates SAP PO #1.

2. Komatsu Sub-Order #2 (EO Emergency) — Ref R230/2026 : 20 units assigned to Machine SN 100433 → Converts to Komatsu SO #0000278047 at confirmed PDX price → Creates SAP PO #2.

3. Komatsu Sub-Order #3 (EO Emergency) — Ref R231/2026 : 20 units assigned to Machine SN 100434 → Converts to Komatsu SO #0000278048 at confirmed PDX price → Creates SAP PO #3.

Each of the 3 resulting SAP Purchase Orders uses the exact same document structure, with the exact price confirmed on the respective Komatsu SO.

5. Step-by-Step Functional Workflow & Screenshots

Step 1: Part Entry, Real-Time Master Stock & Allocation

The maintenance engineer enters requested part numbers. The EQP engine verifies them against the Komatsu catalogue, fetches live warehouse availability badges (**KME Stock** in Dubai, **EOR** restriction, and **KLTD** factory stock), and allocates quantities between available stock and emergency shortage.

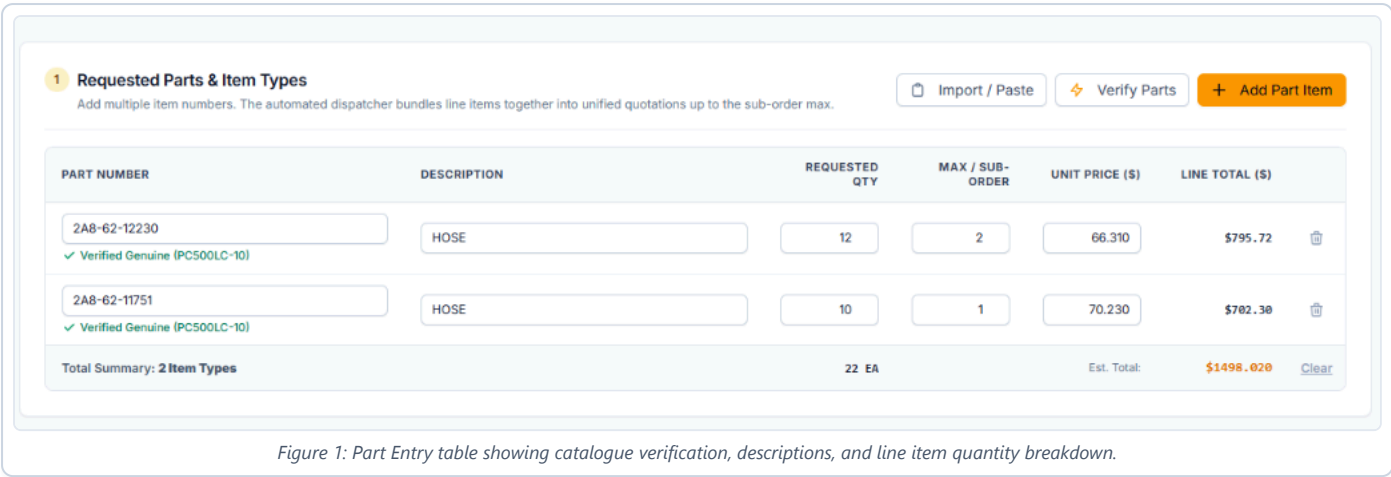


Figure 1: Part Entry table showing catalogue verification, descriptions, and line item quantity breakdown.

Step 2: Fleet Machine Allocation (Emergency Orders Only)

For items routed through the Emergency channel, the dispatcher selects the target customer fleet account (e.g. *Laala Al-Kuwait Real Estate Co.*) and activates compatible machine chassis serial numbers (e.g. *PC500LC-10R*). For 100% Stock orders, machine allocation is completely bypassed.

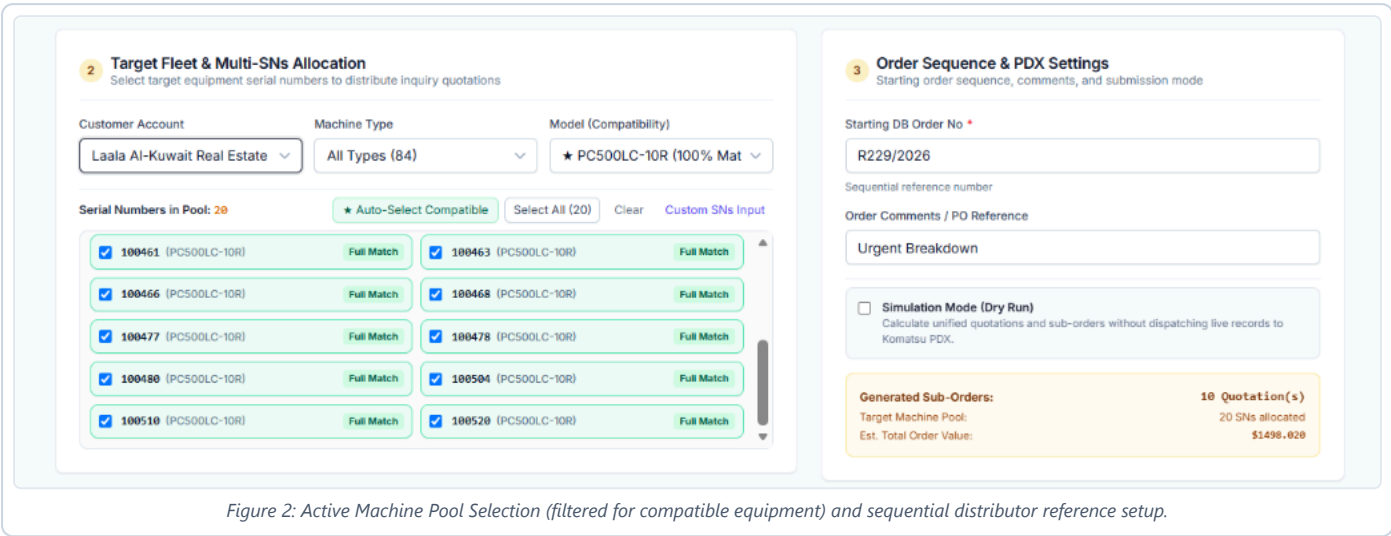


Figure 2: Active Machine Pool Selection (filtered for compatible equipment) and sequential distributor reference setup.

Step 3: Unified Quotations Manifest

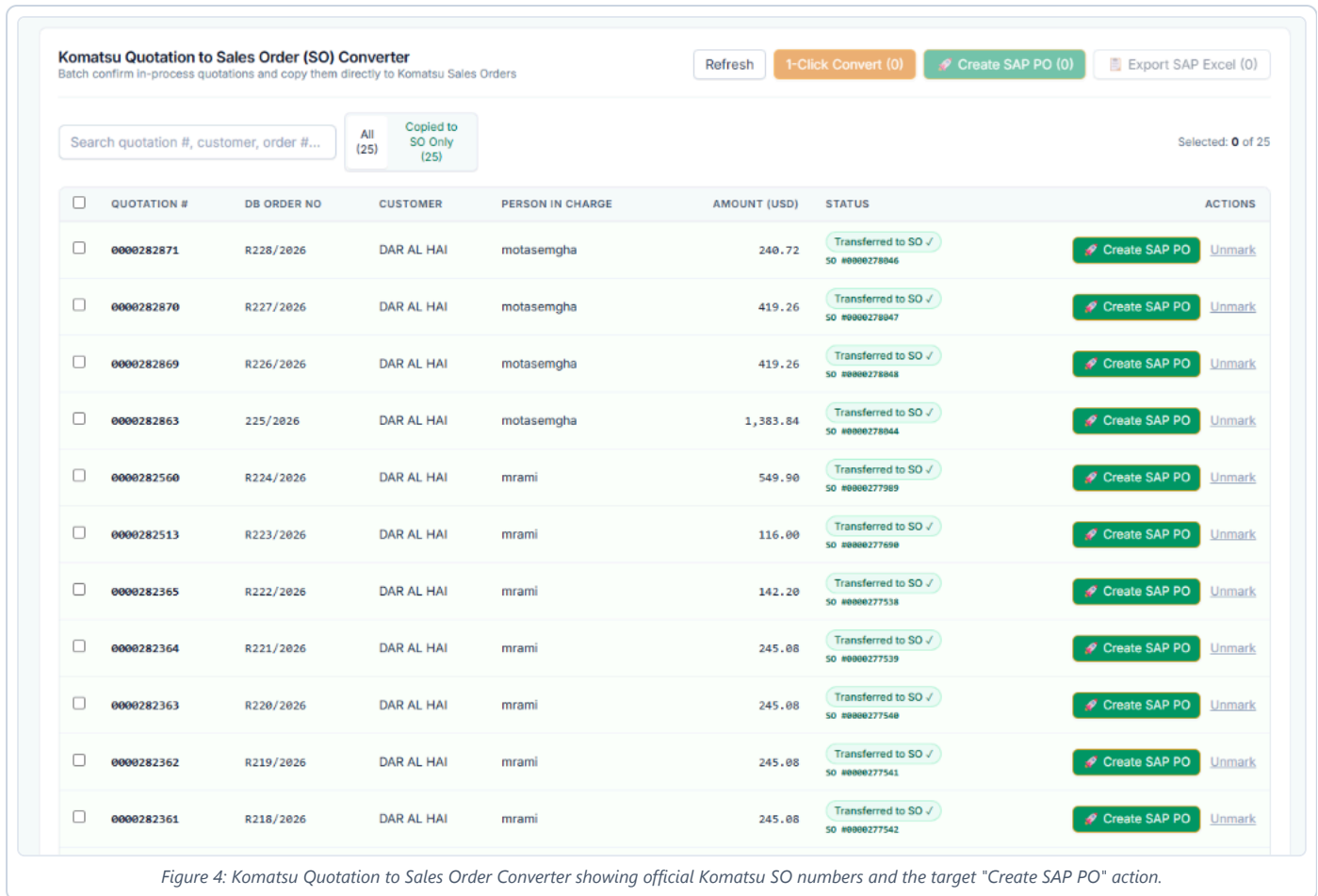
The EQP engine generates the sub-order manifest ready for Komatsu submission, assigning continuous distributor reference numbers (e.g. R229/2026 , R230/2026 , R231/2026) and tracking asset assignments.

4 Planned Unified Quotations & Live Dispatch Queue 10 sub-orders generated across 2 requested parts (22 units) Export CSV Export SAP Excel Dispatch All (10)								
#	DB ORDER NO	TARGET ASSET	BUNDLED PARTS REQUESTED	TOTAL QTY	EST. VALUE (\$)	QUOTATION #	STATUS	ACTION
1	R229/2026	100433 PC500LC-10R • Laala Al-Kuwait Real Estate Co.	2A8-62-12230 : 2 EA 2A8-62-11751 : 1 EA	3	\$202.850	–	Ready	Inspect
2	R230/2026	100434 PC500LC-10R • Laala Al-Kuwait Real Estate Co.	2A8-62-12230 : 2 EA 2A8-62-11751 : 1 EA	3	\$202.850	–	Ready	Inspect
3	R231/2026	100439 PC500LC-10R • Laala Al-Kuwait Real Estate Co.	2A8-62-12230 : 2 EA 2A8-62-11751 : 1 EA	3	\$202.850	–	Ready	Inspect
4	R232/2026	100444 PC500LC-10R • Laala Al-Kuwait Real Estate Co.	2A8-62-12230 : 2 EA 2A8-62-11751 : 1 EA	3	\$202.850	–	Ready	Inspect
5	R233/2026	100447 PC500LC-10R • Laala Al-Kuwait Real Estate Co.	2A8-62-12230 : 2 EA 2A8-62-11751 : 1 EA	3	\$202.850	–	Ready	Inspect
6	R234/2026	100448 PC500LC-10R • Laala Al-Kuwait Real Estate Co.	2A8-62-12230 : 2 EA 2A8-62-11751 : 1 EA	3	\$202.850	–	Ready	Inspect
7	R235/2026	100449 PC500LC-10R • Laala Al-Kuwait Real Estate Co.	2A8-62-11751 : 1 EA	1	\$70.230	–	Ready	Inspect
8	R236/2026	100454 PC500LC-10R • Laala Al-Kuwait Real Estate Co.	2A8-62-11751 : 1 EA	1	\$70.230	–	Ready	Inspect
9	R237/2026	100456 PC500LC-10R • Laala Al-Kuwait Real Estate Co.	2A8-62-11751 : 1 EA	1	\$70.230	–	Ready	Inspect
10	R238/2026	100459 PC500LC-10R • Laala Al-Kuwait Real Estate Co.	2A8-62-11751 : 1 EA	1	\$70.230	–	Ready	Inspect
View Technical Dispatch Audit Logs (4 entries)								

Figure 3: Planned Unified Quotations Manifest showing distributor order references, target assets, and line items.

Step 4: Komatsu Quotation to Sales Order (SO) Conversion & SAP Trigger

Once submitted to Komatsu PDX, the portal returns official Komatsu Quotation Numbers. The EQP system batch-confirms these quotations and converts them into official **Komatsu Sales Orders (SO)** (e.g., SO #0000278046 , SO #0000278047 ...).



The Exact Integration Trigger:

At this stage, the Komatsu Sales Order is officially confirmed. Clicking **"Create SAP PO"** calls the SAP Service Layer API to instantly create the approved Purchase Order in SAP B1 for Vendor **V000006**, returning the SAP DocNum directly into this screen.

6. Technical API Specification for the ABS / SAP Team

To implement direct creation of Purchase Orders in SAP Business One, we propose integrating with the **SAP Business One Service Layer** (OData v4 REST API).

A. Header Field Mapping (Table: `OPOR` / Entity: `PurchaseOrders`)

SAP FIELD	SAP DESCRIPTION	SOURCE FROM EQP SYSTEM	EXAMPLE VALUE
CardCode	Vendor Code	Komatsu Middle East Vendor Code	V000006
DocDate	Posting Date	Order Date (YYYY-MM-DD)	2026-10-05
DocDueDate	Delivery Due Date	Required Delivery Date	2026-10-12
DocCurrency	Document Currency	Komatsu Billed Currency	USD
NumAtCard	Vendor Ref / BP Reference	Komatsu SO Number + DB Order Ref	SO #0000278046 / R229/2026
Comments	Remarks	Order context & reference notes	Komatsu Ref: R229/2026 Quote: 0000282871
U_KomatsuSO	UDF: Komatsu SO No	Official Komatsu Sales Order Number	0000278046
U_MachineSerial	UDF: Machine Serial	Asset Serial (if EO) or blank (if Stock)	100433 (or null)
U_MachineModel	UDF: Machine Model	Equipment Model (if EO) or blank (if Stock)	PC500LC-10R (or null)

B. Line Item Field Mapping (Table: `POR1` / Entity: `DocumentLines`)

SAP FIELD	SAP DESCRIPTION	SOURCE FROM EQP SYSTEM	EXAMPLE VALUE
ItemCode	Item Number	Genuine Komatsu Part Number	600-311-6742
ItemDescription	Item Description	Part Description from Komatsu Master	FUEL FILTER
Quantity	Quantity	Confirmed order line quantity	10.00
UnitPrice	Unit Price (USD)	Confirmed unit price directly from Komatsu SO	25.00
WarehouseCode	Warehouse	Receiving Warehouse	003 (or default per ABS)
TaxCode	Tax Code	Import Zero/Exempt	P0 (or default per ABS)

C. Unified Standard JSON Request Payload (`POST /b1s/v1/PurchaseOrders`)

Because all orders follow the exact same structure in SAP, our backend transmits this uniform payload:

```
{
  "CardCode": "V000006",
  "DocDate": "2026-10-05",
  "DocDueDate": "2026-10-12",
  "DocCurrency": "USD",
  "NumAtCard": "SO #0000278046 / R229/2026",
  "Comments": "Komatsu Order | DB Ref: R229/2026 | Quote: 0000282871",
  "U_KomatsuSO": "0000278046",
  "U_KomatsuQuote": "0000282871",
  "U_MachineSerial": "100433",
  "U_MachineModel": "PC500LC-10R",
  "DocumentLines": [
    {
      "ItemCode": "600-311-6742",
      "ItemDescription": "FUEL FILTER",
      "Quantity": 10.0,
      "UnitPrice": 25.00,
      "WarehouseCode": "003",
      "TaxCode": "P0"
    }
  ]
}
```

D. Expected SAP Response & Storage

Upon successful creation, the SAP B1 Service Layer returns HTTP 201 Created with the official document identifiers:

```
{
  "DocEntry": 14205,
  "DocNum": 20260481,
  "DocDate": "2026-10-05",
  "CardCode": "V000006",
  "DocTotal": 250.00,
  "DocCurrency": "USD"
}
```

The EQP database stores `DocNum: 20260481` and `DocEntry: 14205` directly onto the Komatsu quotation record, giving maintenance, procurement, and accounting teams complete cross-traceability.

7. Key Questions & Alignment Needed from the ABS Team

1. Service Layer Endpoint URL & Connectivity:

Please provide the official Service Layer URL (e.g. `https://daralhai.b1pro.com:50000/b1s/v1/`) and confirm firewall whitelisting for the EQP backend server.

2. Dedicated Technical API User:

We request a dedicated API service account (e.g. `B1_API_EQP`) with permissions restricted to `PurchaseOrders` (Create/Read) and `Items` (Read/Create) to avoid using individual user credentials.

3. Handling New Part Numbers (Item Master `OITM`):

With over 500,000 Komatsu parts in the catalogue, occasionally a component does not yet exist in SAP B1. What is ABS's preferred procedure?

- *Option 1 (Recommended):* Allow our API to auto-create missing parts via `POST /b1s/v1/Items` using Komatsu catalogue master data before creating the PO.
- *Option 2:* Use a standard non-inventory item master with explicit line-item descriptions.

4. Document Status: Active PO vs. Draft:

Should the API create approved open POs (`PurchaseOrders`) directly, or create Drafts (`PurchaseOrderDrafts`) for accounting verification during the initial pilot phase?

5. Receiving Warehouse Code:

Please confirm the default receiving warehouse code for Komatsu parts (e.g. `003` or `001`).

6. Existing UDF Confirmation:

Please confirm if user-defined fields already exist on `OPOR` for `U_KomatsuSO`, `U_MachineSerial`, and `U_MachineModel`, or if ABS will create them.

Next Steps & Testing Plan:

Once ABS provides the Service Layer connection credentials and confirms the item master procedure, the EQP engineering team will execute sample test transactions in the SAP sandbox environment within 24 hours.